USB-C input and 3.3 V supply

File: power.kicad_sch

RP2350B microcontroller

File: mcu.kicad_sch

Chip breakout sockets and I/O

File: breakout.kicad_sch

Chip core power monitor

File: power_monitor.kicad_sch

Host board for the sparsity-aware convolution chip (Tiny Tapeout, SkyWater 130 nm).
Keeps the Tiny Tapeout demo board v3 RP2350B pin map, so its firmware and the host/ test kit run unchanged,
and adds an INA228 power monitor on the chip core supply. Demo board design (c) 2025 Pat Deegan, Apache 2.0.

Top level: one block per sheet

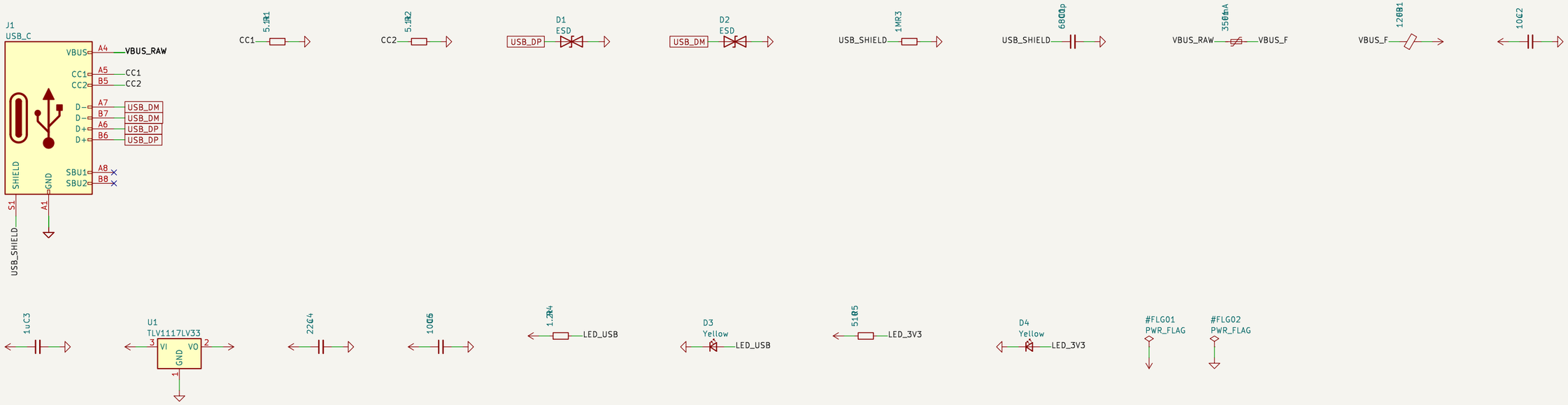
Izaiah Thigpen

Sheet: /

File: sparse-conv-host.kicad_sch

Title: Sparse Convolution Chip Host Board

Size: A4	Date: 2026-10-22	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/5



USB-C, polyfuse, ESD and the 3.3 V LDO are as on the Tiny Tapeout demo board v3.
The LDO also feeds the breakout (which makes the chip 1.8 V core) and the power monitor.

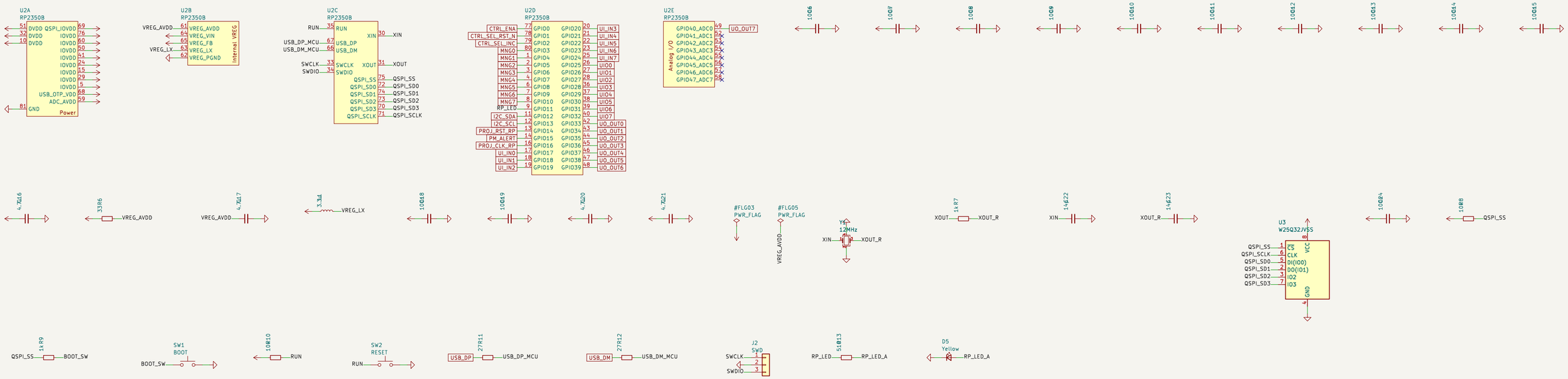
Power

Izaiah Thigpen

Sheet: /USB-C input and 3.3 V supply/
File: power.kicad_sch

Title: USB-C input and 3.3 V supply

Size: A3	Date: 2026-10-22	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/5



Copied from the Tiny Tapeout demo board v3 RP2350B sheet. GPIO map (unchanged, so the TT firmware runs as is):
GPIO0 ctrl_ena, GPIO1 ctrl_sel_rst_n, GPIO2 ctrl_sel_inc, GPIO3–10 management (breakout ID EEPROM on GPIO4/5),
GPIO11 LED, GPIO14 project reset, GPIO16 project clock, GPIO17–24 ui_in, GPIO25–32 uio, GPIO33–40 uo_out.
New: GPIO12/13 I2C to the INA228 power monitor, GPIO15 its ALERT.

MCU
Izaiah Thigpen
Sheet: /RP2350B microcontroller/
File: mcu.kicad_sch

Title: RP2350B microcontroller

Size: A2	Date: 2026-10-22	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/5

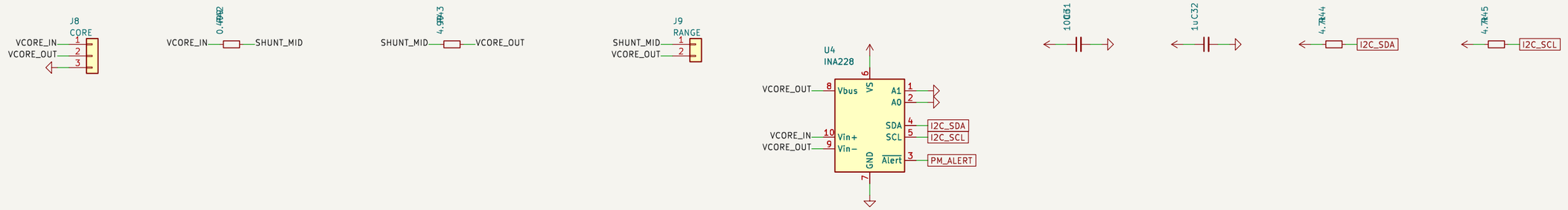


The breakout socket, pull-ups and PMODs follow the Tiny Tapeout demo board v3 exactly.
Logic-analyzer header for this design: ui_in[0..3] = SCLK, CS_N, MOSI, BYPASS;
uo_out[0..3] = MISO, DONE, SKIP, BUSY; uo_out[4..7] = TAP[0..3].
Analog pins: 10 Mohm pull-downs only, as on the demo board. RES pins left open.

Breakout
Izaiah Thigpen
Sheet: /Chip breakout sockets and I/O/
File: breakout.kicad_sch

Title: Chip breakout sockets and I/O

Size: A2	Date: 2026-10-22	**Rev: A**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 4/5



Core current = V(shunt) / R. Normal range: 0.499 + 4.99 ohm = 5.49 ohm, 7.5 mA full scale (ADCRANGE=1).
High range: fit a jumper on RANGE to short the 4.99 ohm -> 0.499 ohm, 82 mA full scale.
Hook-up: on the TT breakout remove R14 (0 ohm, 0805), fit header J2, then wire
J2 pin 5 (VCC_REG) -> CORE pin 1, J2 pin 6 (VCCD) -> CORE pin 2, ground -> CORE pin 3.
Layout: Kelvin traces from the shunt pads to IN+/IN-; INA228 address 0x40 (A0 = A1 = GND).

PowerMonitor
Izaiah Thigpen
Sheet: /Chip core power monitor/
File: power_monitor.kicad_sch

Title: Chip core power monitor

Size: A3	Date: 2026-10-22	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/5